animals release carbon into atmosphere in the form of carbon dioxide. Carbon dioxide is also returned to the atmosphere through decomposition of dead organic matter, burning fossil fuels and volcanic activities.

Human impacts on carbon cycle

More carbon moves into the atmosphere due to burning of fossil fuels and deforestation. Most of the carbon in atmosphere is in the form of carbon dioxide. Carbon dioxide is a greenhouse gas. By increasing the amount of carbon dioxide, earth becomes warmer. This leads to greenhouse effect and global warming.

24.2 Adaptations of Plants

Any feature of an organism or its part that enables it to exist under conditions of its habitat is called adaptation. On the basis of water availability, plants have been classified as:

(i) Hydrophytes
(ii) Xerophytes
(iii) Mesophytes



24.2.1 Hydrophytes

Plants growing in or near water are called hydrophytes. Hydrophytes may be free floating or submerged plants living in lakes, ponds, shallow water, marshy lands and marine habitat. Hydrophytes face certain challenges in their habitat. They are:

(i) Availability of more water than needed.
(ii) Water current may damage the plant body.
(iii) Water levels may change regularly.
(iv) Maintain buoyancy in water.

Adaptations of hydrophytes

1. Roots are poorly developed as in *Hydrilla* or absent as in *Wolffia*.
2. Plant body is greatly reduced as in *Lemna*.
3. Submerged leaves are narrow or finely divided. e.g. *Hydrilla*.


Hydrilla


Wolffia


Eichhornia


Lemna

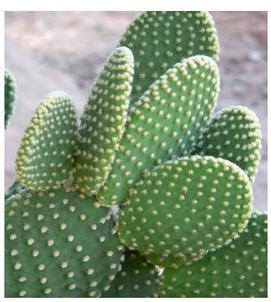
Figure 24.4 Hydrophytes

4. Floating leaves have long leaf stalks to enable the leaves move up and down in response to changes in water level. e.g. Lotus.
5. Air chambers provide buoyancy and mechanical support to plants as in *Eichhornia* (swollen and spongy petiole).

24.2.2 Xerophytes

Plants that grow in dry habitat are called xerophytes. These plants develop special structural and physiological characteristics to meet the following conditions:

(i) To absorb as much water as they can get from the surroundings.
(ii) To retain water in their organs for very long time.
(iii) To reduce the transpiration rate.
(iv) To reduce consumption of water.


Opuntia


Calotropis

Figure 24.5 Xerophytes

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